Aim:

To efficiently detect and display duplicate alphanumeric strings within a fixed-size dataset by sorting the data in memory and performing a single-pass adjacent comparison.

Algorithm

1. Initialize an array of 5 elements and populate it with sample alphanumeric strings.
2. Set a control flag (WS-DUPLICATE-FOUND) to 'N' to track duplicate occurrences.
3. Sort the array in ascending alphanumeric order to group identical strings side-by-side.
4. Iterate through the array from the first element up to the second-to-last element.
5. During each iteration, calculate the next index position and compare the current string with the immediate next string.
6. If a match occurs, print the duplicate string with its index positions and update the control flag to 'Y'.
7. After the loop completes, check the control flag; if it remains 'N', print a message stating no duplicates exist.
8. Terminate the program execution.

Program:

```
IDENTIFICATION DIVISION.
    PROGRAM-ID. CHECK-DUPLICATES-SORT.

    DATA DIVISION.
    WORKING-STORAGE SECTION.

    01  WS-TABLE-DATA.
       05  WS-ITEM            PIC X(10) OCCURS 5 TIMES.

    01  WS-TOTAL-COUNT         PIC 9(3) VALUE 5.
    01  WS-I                   PIC 9(3) VALUE 0.
    01  WS-NEXT                PIC 9(3) VALUE 0.
    01  WS-DUPLICATE-FOUND     PIC X VALUE 'N'.

    PROCEDURE DIVISION.
    MAIN-LOGIC.
       MOVE "XYZ789AB" TO WS-ITEM(1)
       MOVE "ABC123XY" TO WS-ITEM(2)
       MOVE "DEF456ZZ" TO WS-ITEM(3)
       MOVE "ABC123XY" TO WS-ITEM(4)
       MOVE "GHI012JK" TO WS-ITEM(5)

       *> Correct syntax: table name without (ALL)
```

```
SORT WS-ITEM ASCENDING KEY WS-ITEM

PERFORM VARYING WS-I FROM 1 BY 1
      UNTIL WS-I >= WS-TOTAL-COUNT
   COMPUTE WS-NEXT = WS-I + 1
   IF WS-ITEM(WS-I) = WS-ITEM(WS-NEXT)
      MOVE 'Y' TO WS-DUPLICATE-FOUND
      DISPLAY "Duplicate Detected: " WS-ITEM(WS-I)
            " (Matched at sorted positions "
            WS-I " and " WS-NEXT ")"
   END-IF
END-PERFORM

IF WS-DUPLICATE-FOUND = 'N'
   DISPLAY "No duplicate strings found."
END-IF

STOP RUN.
```

Output:

```
Output:
Duplicate Detected: ABC123XY   (Matched at sorted positions 001 and 002)
Duplicate Detected: XYZ789AB   (Matched at sorted positions 005 and 006)
```